1) & 3) boundary value testing ,sales com

```
#include<stdio.h>

#include<conio.h>

void main()

{

   int locks, stocks, barrels;

   int tot_l = 0, tot_s = 0, tot_b = 0;

   float sales, com;

   printf("Enter locks (-1 to stop): ");

   scanf("%d", &locks);

   while (locks != -1)

   {

      printf("Enter stocks and barrels: ");

      scanf("%d %d", &stocks, &barrels);

      // Validation

      if (locks < 1 || locks > 70)

         printf("Invalid locks\n");

      else if (stocks < 1 || stocks > 80)

         printf("Invalid stocks\n");

      else if (barrels < 1 || barrels > 90)

         printf("Invalid barrels\n");

      else

      {
```

```
        tot_l += locks;
        tot_s += stocks;
        tot_b += barrels;
    }
    printf("\nEnter locks (-1 to stop): ");
    scanf("%d", &locks);
}
// Sales calculation
sales = tot_l * 45 + tot_s * 30 + tot_b * 25;
// Commission (ORIGINAL STYLE)
if (sales <= 1000)
{
    com = 0.10 * sales;
}
else if (sales <= 1800)
{
    com = 0.10 * 1000;
    com = com + (0.15 * (sales - 1000));
}
else
{
    com = 0.10 * 1000;
    com = com + (0.15 * 800);
```

```
        com = com + (0.20 * (sales - 1800));
    }

    printf("\nTotal Locks = %d", tot_l);
    printf("\nTotal Stocks = %d", tot_s);
    printf("\nTotal Barrels = %d", tot_b);
    printf("\nTotal Sales = %.2f", sales);
    printf("\nCommission = %.2f", com);
    getch();
}
```

2) & 4) the Next Date function. Analyze it from the perspective of boundary value testing and Equi  testing.

```
#include<stdio.h>
#include<conio.h>
main( )
{
int
month[12]={31,28,31,30,31,30,31,31,30,31,30,31};
int d,m,y,nd,nm,ny,ndays;
printf("enter the date,month,year");
scanf("%d%d%d",&d,&m,&y);
```

```
ndays=month[m-1];
if(y<=1812 && y>2012)
{
printf("Invalid Input Year");
exit(0);
}
if(d<=0 | d>ndays)
{
printf("Invalid Input Day");
exit(0);
}
if(m<1 && m>12)
{
printf("Invalid Input Month");
exit(0);
}
if(m==2)
{if(y%100==0)
{if(y%400==0)
ndays=29;
}
else if(y%4==0)
ndays=29;
```

```
}
nd=d+1;
nm=m; ny=y;
if(nd>ndays)
{
nd=1;
nm++;
}
if(nm>12)
{
nm=1;
ny++;
}
printf("\n Given date is %d:%d:%d",d,m,y);
printf("\n Next day’s date is %d:%d:%d",nd,nm,ny);
getch( );
}
```

5. implement triangle problem

```
#include <stdio.h>
#include<conio.h>
int main()
```

```
{
int a,b,c,c1,c2,c3;
do {
printf("Enter sides of triangle:\n");
scanf("%d %d %d",&a,&b,&c);
c1=((a>=1) && (a<=10));
c2=((b>=1) && (b<=10));
c3=((c>=1) && (c<=10));
if(!c1)
printf("Value of a is out of range:");
if(!c2)
printf("Value of b is out of range:");
if(!c3)
printf("Value of b is out of range:");
}
while(!c1 ||!c2 ||!c3);
if((a+b)>c && (b+c)>a && (c+a)>b)
{
if(a==b && b==c)
printf("Triangle is Equilateral \n");
else if(a!=b && b!=c && c!=a)
printf("Triangle is Scalar \n");
else
```

```
printf("Triangle is Isosceles \n");
}
else("Triangle Can’t be formed \n");
getch();
return 0;
}
```